Answers and working

Mark your work, then fix each mistake in a different colour. Rounded answers are to 1 decimal place unless the question says otherwise.

Topic 1 · Your turn 1 — Squares, powers and square roots (page 1)

a $6 \times 6 = 36$

b $11 \times 11 = 121$

c $0.3 \times 0.3 = 0.09$

d $3 \times 3 \times 3 \times 3 = 81$

e $10 \times 10 \times 10 = 1000$

f 9, because $9^2 = 81$

g 14, because $14^2 = 196$

h $25 + 144 = 169$

i $\sqrt{169} = 13$

Topic 1 · Your turn 2 — The hypotenuse (page 1)

a *PR* (opposite the right angle at *Q*)

b *XY* (opposite the right angle at *Z*)

c *MN* (opposite the right angle at *L*)

d $r^2 = p^2 + q^2$ (r is opposite the right angle)

Topic 1 · Your turn 3 — Finding the hypotenuse (page 1)

a $x^2 = 6^2 + 8^2 = 36 + 64 = 100$, $x = \sqrt{100} = 10$ cm

b $x^2 = 20^2 + 21^2 = 400 + 441 = 841$, $x = \sqrt{841} = 29$ m

c $x^2 = 7^2 + 10^2 = 49 + 100 = 149$, $x = \sqrt{149} \approx 12.2$ cm

d $x^2 = 2.4^2 + 1.8^2 = 5.76 + 3.24 = 9$, $x = \sqrt{9} = 3$ m

Topic 1 · Your turn 4 — Finding a shorter side (page 2)

a $y^2 = 10^2 - 8^2 = 100 - 64 = 36$, $y = \sqrt{36} = 6$ cm

b $y^2 = 25^2 - 15^2 = 625 - 225 = 400$, $y = \sqrt{400} = 20$ m

c $y^2 = 9^2 - 5^2 = 81 - 25 = 56$, $y = \sqrt{56} \approx 7.5$ cm

d $x^2 = 6^2 + 2.5^2 = 36 + 6.25 = 42.25$, $x = \sqrt{42.25} = 6.5$ m

Topic 1 · Your turn 5 — Surds (page 2)

a $\sqrt{8}$, $\sqrt{30}$, $\sqrt{45}$, $\sqrt{2}$ (the others are 1, 7, 10 and 12)

b 5 and 6, because $5^2 = 25$ and $6^2 = 36$, and 30 is between 25 and 36.

c $x^2 = 3^2 + 5^2 = 9 + 25 = 34$, $x = \sqrt{34}$ (exact) ≈ 5.8 cm

Topic 1 · Your turn 6 — Pythagoras in real life (page 2)

a $d^2 = 40^2 + 30^2 = 1600 + 900 = 2500$, $d = \sqrt{2500} = 50$ m. The path is 50 m long.

b $d^2 = 7^2 + 15^2 = 49 + 225 = 274$, $d = \sqrt{274} \approx 16.6$ cm

c $h^2 = 30^2 - 18^2 = 900 - 324 = 576$, $h = \sqrt{576} = 24$ m. The kite is 24 m above her hand.

Topic 1 · Your turn 7 — Is it right-angled? (page 3)

a $10^2 = 100$; $6^2 + 8^2 = 36 + 64 = 100$. **Equal, so it is right-angled.**

b $6^2 = 36$; $4^2 + 5^2 = 16 + 25 = 41$. **$36 \neq 41$, so it is not right-angled.**

c $41^2 = 1681$; $9^2 + 40^2 = 81 + 1600 = 1681$. **Equal, so it is right-angled.**

d $15^2 = 225$; $10^2 + 11^2 = 100 + 121 = 221$. **$225 \neq 221$, so it is not right-angled.**

e $2.5^2 = 6.25$; $1.5^2 + 2^2 = 2.25 + 4 = 6.25$. **Equal, so it is right-angled.**

f $61^2 = 3721$; $11^2 + 60^2 = 121 + 3600 = 3721$. **Equal, so it is right-angled.**

Topic 1 · Your turn 8 — Pythagorean triads (page 3)

a $5^2 + 12^2 = 25 + 144 = 169 = 13^2$ ✓

b 15, 20, 25: $15^2 + 20^2 = 225 + 400 = 625 = 25^2$ ✓

c 9, 12, 15 ($81 + 144 = 225 = 15^2$); 10, 24, 26 ($676 - 100 = 576 = 24^2$)

Topic 1 · Your turn 9 — Exam-style practice (page 3)

1 $x^2 = 12^2 + 35^2 = 144 + 1225 = 1369$, $x = \sqrt{1369} = 37$ cm

2 $13^2 = 169$; $8^2 + 10^2 = 64 + 100 = 164$. **$169 \neq 164$, so it is not right-angled.**

3 1 m = 100 cm. $100^2 = 10000$; $60^2 + 80^2 = 3600 + 6400 = 10000$. **Equal, so it is right-angled.**

4 Half the base is 6 cm. $h^2 = 10^2 - 6^2 = 100 - 36 = 64$, $h = \sqrt{64} = 8$ cm

Topic 1 · Extension — Simplifying surds (page 3)

a $\sqrt{8} = \sqrt{4 \times 2} = 2\sqrt{2}$

b $\sqrt{20} = \sqrt{4 \times 5} = 2\sqrt{5}$

c $\sqrt{27} = \sqrt{9 \times 3} = 3\sqrt{3}$

d $\sqrt{75} = \sqrt{25 \times 3} = 5\sqrt{3}$

Topic 2 · Your turn 1 — Equivalent fractions and mixed numerals (page 4)

a $\frac{8}{12}$ (multiply top and bottom by 4)

b $\frac{3}{5}$ ($\div 5$)

c $\frac{3}{4}$ ($\div 8$)

d $\frac{3 \times 7 + 2}{7} = \frac{23}{7}$

e $29 \div 6 = 4$ remainder 5, so $4\frac{5}{6}$

f $\frac{16}{24}, \frac{18}{24}, \frac{15}{24} \rightarrow \frac{5}{8}, \frac{2}{3}, \frac{3}{4}$

Topic 2 · Your turn 2 — Adding and subtracting fractions (page 4)

a $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

b $\frac{8}{10} - \frac{5}{10} = \frac{3}{10}$

c $\frac{9}{12} + \frac{10}{12} = \frac{19}{12} = 1\frac{7}{12}$

d $\frac{9}{4} + \frac{5}{3} = \frac{27}{12} + \frac{20}{12} = \frac{47}{12} = 3\frac{11}{12}$

e $\frac{40}{8} - \frac{19}{8} = \frac{21}{8} = 2\frac{5}{8}$

Topic 2 · Your turn 3 — Multiplying and dividing fractions (page 4)

a $= \frac{6}{35}$

b $\frac{1}{2} \times \frac{1}{2}$ (cancel 5 and 10, 3 and 6) $= \frac{1}{4}$

c $\frac{4}{3} \times \frac{3}{8} = \frac{1}{2}$

d $\frac{3}{4} \times \frac{2}{1} = \frac{6}{4} = 1\frac{1}{2}$

e $\frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$

f $\frac{7}{2} \times \frac{4}{7} = 2$

g black $\frac{1}{3} \times 36 = 12$, brown $\frac{1}{4} \times 36 = 9$, white $36 - 12 - 9 = 15$

h $\frac{9}{4} \times 3 = \frac{27}{4} = 6\frac{3}{4}$ cups

Topic 2 · Your turn 4 — Fractions and decimals, rounding (page 5)

a $\frac{8}{10} = \frac{4}{5}$

b $\frac{45}{100} = \frac{9}{20}$

c $\frac{125}{1000} = \frac{1}{8}$

d $3 \div 5 = 0.6$

e $7 \div 20 = 0.35$

f $5 \div 6 = 0.8\dot{3}$

g $4 \div 9 = 0.\dot{4}$

h $3 \div 11 = 0.\dot{2}\dot{7}$

i 5.4 (next digit 2 → round down)

j 12.40 (next digit 5 → round up; keep the 0)

Topic 2 · Your turn 5 — Multiplying decimals (page 5)

a $4 \times 3 = 12$; 2 decimal places → 0.12

b $9 \times 6 = 54$; 3 decimal places → 0.054

c $25 \times 8 = 200$; 2 decimal places → 2

d $16 \times 45 = 720$; 3 decimal places → 0.72

e $32 \times 15 = 480$; 3 decimal places → 0.48

f $75 \times 12 = 900$; 3 decimal places → 0.9

Topic 2 · Your turn 6 — Dividing decimals (page 5)

a × 10: $36 \div 4 = 9$

b × 100: $144 \div 12 = 12$

c × 100: $810 \div 9 = 90$

d × 100: $56 \div 8 = 7$

e × 10: $150 \div 5 = 30$

f × 10: $94.5 \div 3 = 31.5$

Topic 2 · Your turn 7 — Decimals in real life (page 5)

a $12 \times \$14.90 = \$178.80$; $8 \times \$4.75 = \$38.00$; total \$216.80

b $\$4.20 \div 1.5 = 42 \div 15 = \$2.80$ per kg

c $3 \times \$2.85 = \$8.55$; $\$8.55 + \$8.40 = \$16.95$; $\$50 - \$16.95 = \$33.05$

Topic 2 · Your turn 8 — Fractions, decimals and percentages (page 6)

a $\frac{1}{4} = 0.25 = 25\%$

b $\frac{3}{5} = 0.6 = 60\%$

c $\frac{3}{25} = 0.12 = 12\%$

d $\frac{7}{8} = 0.875 = 87.5\%$

e $\frac{1}{40} = 0.025 = 2.5\%$

Topic 2 · Your turn 9 — Percentage of a quantity (page 6)

a $70 \div 10 = 7$

b $36 \div 4 = 9$

c $0.4 \times 85 = 34$

d $0.12 \times 250 = \$30.00$

e $64 \div 8 \times 3 = 24$

f 35% of 160 $= 0.35 \times 160 = 56$ students

g Walk 20% of 160 $= 32$; bike 10% of 160 $= 16$; $32 - 16 = 16$ more

Topic 2 · Your turn 10 — One quantity as a percentage of another (page 6)

a $\frac{34}{40} \times 100\% = 85\%$

b $\frac{12}{30} \times 100\% = 40\%$

c 2 h = 120 min; $\frac{30}{120} \times 100\% = 25\%$

Topic 2 · Your turn 11 — Percentage increase and decrease (page 6)

a 10% of 60 $= 6$; $60 + 6 = \$66.00$

b 20% of 250 $= 50$; $250 - 50 = 200$ kg

c 30% of 45 $= 13.50$; $45 - 13.50 = \$31.50$

d 1% $= 54 \div 30 = 1.8$; $100\% = 1.8 \times 100 = 180$ students

Topic 3 · Your turn 1 — The language of algebra (page 7)

a (i) 6 (ii) −1 (iii) 4 (iv) 3

b $5y$

c $3ab$

d m^2

e $2p^2$

f $\frac{k}{6}$

g $4 \times m \times n$

h $3 \times c \times c$

Topic 3 · Your turn 2 — Words to algebra (page 7)

a $n + 7$ **b** $4n$ **c** $n - 3$

d $\frac{n}{5}$ **e** $2n + 1$ **f** $n + 1$

g $3(x + y)$

Topic 3 · Your turn 3 — Substitution (page 7)

a $3 \times (5) + (2) = 15 + 2 = 17$

b $(5) \times (2) - 4 = 10 - 4 = 6$

c $(5)^2 + (2)^2 = 25 + 4 = 29$

d $2 \times (2)^2 = 2 \times 4 = 8$ (square first)

e $3 \times 25 - 4 \times 2 = 75 - 8 = 67$

f $(5) + (-3) = 2$

g $(-3)^2 = (-3) \times (-3) = 9$

h $4 \times (-3) - (2) = -12 - 2 = -14$

i $(5 + 2)^2 = 7^2 = 49$

j (i) $45 \times 3 + 60 = \$195.00$ (ii) $45 \times 2.5 + 60 = \$172.50$

Topic 3 · Your turn 4 — Collecting like terms (page 8)

a $8a$

b $8k$

c $5p + 4$

d $7x + 2y$

e $3p + 3q$

f $4ab$

g $-x$ and $7x$

h $7y^2 + 3y$

i $-3m - 3$

j $(3x + 2) + x + (3x + 2) + x = 8x + 4$

Topic 3 · Your turn 5 — Multiplying terms (page 8)

a $15k$ **b** $14pq$ **c** $12y^2$

d $-12ab$ **e** $30mn$ **f** $12x^2$

Topic 3 · Your turn 6 — Dividing terms (page 8)

a $3m$ **b** $3x$ **c** $3b$

d $2k$ **e** $-5m$

Topic 3 · Your turn 7 — Reasoning with algebra (page 8)

a $3x + 2y = 6 + 2 = 8$, but $5xy = 5 \times 2 \times 1 = 10$. $8 \neq 10$, so Sam is wrong: $3x$ and $2y$ are not like terms, so they cannot be added.

b $4m$

c $3a$

d $5x \times 3x = 15x^2$

e $4a^2 = 4 \times 9 = 36$; $(4a)^2 = 12^2 = 144$. Not the same: in $4a^2$ only a is squared.

Topic 3 · Your turn 8 — Expanding brackets (page 9)

a $6m + 12$ **b** $12k - 15$ **c** $-2y - 16$

d $a^2 - 9a$ **e** $8x + 7$ $(8x + 4 + 3)$ **f** $7p + 4$ $(5p + 10 + 2p - 6)$

Topic 3 · Your turn 9 — Factorising (page 9)

a $4(x + 2)$ **b** $3(3m - 2)$ **c** $6(2y + 3)$

d $k(k - 7)$ **e** $7p(2q + 3)$

Mixed exam check — all three topics (page 9)

1 $5a - 4b$

2 $12m - 8$

3 $3 \times (2)^2 - 2 \times (-5) = 12 + 10 = 22$

4 $x^2 = 16^2 + 30^2 = 256 + 900 = 1156$, $x = \sqrt{1156} = 34$ cm

5 $7 \times 8 = 56$; 3 decimal places → 0.056

6 × 100: $240 \div 3 = 80$

7 $0.45 \times 360 = 162$

8 $\frac{5}{2} + \frac{3}{4} = \frac{10}{4} + \frac{3}{4} = \frac{13}{4} = 3\frac{1}{4}$

9 $12^2 = 144$; $7^2 + 9^2 = 49 + 81 = 130$. **144 ≠ 130, so it is not right-angled.**

10 8% of 150 = 12; 150 + 12 = \$162.00